```
labex:project/ $ whoami
labex
labex:project/ $ id
uid=5000(labex) gid=5000(labex) groups=5000(labex),27(sudo),121(ssl-cert),5002(public)
labex:project/ $ sudo useradd -m -U -s /bin/bash jack
labex:project/ $ id jack
uid=5002(jack) gid=5003(jack) groups=5003(jack)
labex:project/ $ ls -ld /home/jack
drwxr-x--- 2 jack jack 57 Nov  3 01:44 /home/jack
labex:project/ $ whoami
```

```
labex:project/ $ touch /home/labex/project/secure-report.txt
labex:project/ $ sudo chown jack:developers /home/labex/project/secure-report.txt
labex:project/ $ sudo chmod 640 /home/labex/project/secure-report.txt
labex:project/ $ ls -l /home/labex/project/secure-report.txt
-rw-r----- 1 jack developers 0 Nov  3 01:49 /home/labex/project/secure-report.txt
labex:project/ $ stat -c 'owner=%U group=%G mode=%a' /home/labex/project/secure-report.txt
owner=jack group=developers mode=640
labex:project/ $
```

```
labex:project/ $ sudo getent shadow contractor | cut -d: -f2 | cut -c1
$
labex:project/ $ sudo usermod -L contractor
labex:project/ $ sudo getent shadow contractor | cut -d: -f2 | cut -c1
!
labex:project/ $ sudo usermod -U contractor
labex:project/ $ sudo getent shadow contractor | cut -d: -f2 | cut -c1
$
labex:project/ $ sudo usermod -L contractor
labex:project/ $ ls -l /home/labex/project/secure-report.txt
```

```
labex:project/ $ mkdir -p /home/labex/project/umask-demo
labex:project/ $ original_umask=$(umask)
labex:project/ $ umask 027
labex:project/ $ touch /home/labex/project/umask-demo/private-note.txt
labex:project/ $ mkdir /home/labex/project/umask-demo/private-dir
labex:project/ $ stat -c '%a %n' /home/labex/project/umask-demo/private-note.txt /home/labex/project/umask-demo/private-dir
640 /home/labex/project/umask-demo/private-note.txt
750 /home/labex/project/umask-demo/private-dir
labex:project/ $ umask "$original_umask"
labex:project/ $
```

```
labex:project/ $ mkdir /tmp/shared-dropbox
labex:project/ $ chmod 1777 /tmp/shared-dropbox
labex:project/ $ sudo -u jack touch /tmp/shared-dropbox/jack-note.txt
labex:project/ $ sudo -u contractor touch /tmp/shared-dropbox/contractor-note.txt
labex:project/ $ sudo -u jack rm /tmp/shared-dropbox/contractor-note.txt
rm: remove write-protected regular empty file '/tmp/shared-dropbox/contractor-note.txt'? y
rm: cannot remove '/tmp/shared-dropbox/contractor-note.txt': Operation not permitted
labex:project/ $ ls -ld /tmp/shared-dropbox
drwxrwxrwt 2 labex labex 54 Nov  3 01:54 /tmp/shared-dropbox
labex:project/ $
```

```
labex:project/ $ mkdir /home/labex/project/team-share
labex:project/ $ sudo chown jack:developers /home/labex/project/team-share
labex:project/ $ sudo chmod 2770 /home/labex/project/team-share
labex:project/ $ ls -ld /home/labex/project/team-share
drwxrws--- 2 jack developers 6 Nov  3 01:55 /home/labex/project/team-share
labex:project/ $ stat -c 'owner=%U group=%G mode=%a' /home/labex/project/team-share
owner=jack group=developers mode=2770
labex:project/ $
```

```
labex:project/ $ newgrp developers
labex:project/ $ id
uid=5000(labex) gid=5004(developers) groups=5004(developers),27(sudo),121(ssl-cert),5000(labex),5002(public)
labex:project/ $ touch /home/labex/project/team-share/labex-note.txt
labex:project/ $ stat -c 'owner=%U group=%G mode=%a' /home/labex/project/team-share/labex-note.txt
owner=labex group=developers mode=664
labex:project/ $ exit
labex:project/ $
```



Labby

Summary

You practiced the core identity and access model used by Linux systems. You created a user and team group, managed supplementary memberships, assigned ownership, and translated numeric permission modes into practical access rules. You also inspected an account lock without exposing its password hash.

You then moved beyond ordinary permissions: `umask` controlled defaults, the sticky bit protected files in a public directory, and setgid preserved team group ownership in a private collaboration directory. Finally, you used `newgrp` to refresh group membership and verified the inherited group on a real file.

Continue